A Meta-Analysis of Parent-Involved Treatment for Child Sexual Abuse

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Sexual abuse in children not only occurs with alarming frequency, it also potentially leads to deleterious consequences for victims. Previous narrative reviews have touted the benefits of including the nonoffending caregiver in child sexual treatment. Objective: A meta-analysis is conducted to determine the effects of parent-involved treatment in four major child symptom areas: internalizing, externalizing, sexualized behaviors, and posttraumatic stress. Method: Only comparison/control group studies are included. Participant, treatment, and methodological characteristics of studies are coded, as well as information to calculate effect sizes. An overall effect size is calculated, though the small number of studies precluded moderator analysis. Results: Seven studies are located that met the inclusion criteria. At posttest, treatment had small effects in the four outcome domains over alternative interventions and was reduced still further at follow-up. Conclusion: Parent-involved treatment confers some advantage over comparison conditions (typically child-only treatment).

Keywords: *sexual abuse; treatment outcome; posttraumatic stress disorder; meta-analysis*

Rates of sexual abuse in North America range from 30% to 40% of females and 13% of males (Bolen & Scannapieco, 1999; Briere & Elliott, 2003). Along with these high prevalence rates are the potentially serious effects of sexual abuse, which may include short-term effects of depression, suicide, sexual promiscuity, sexual perpetration, and academic underachievement (Paolucci, Genuis, & Violato, 2001). The long-term impact of child sexual abuse in adulthood may involve depression, low self-esteem, somatization, anger, posttraumatic stress symptomatology, anxiety, obsessions and compulsions, sexual dysfunctions, self-mutilation, suicidality, dissociation, substance abuse problems, revictimization, and relationship problems (Jumper, 1995; Neumann, Houskamp, Pollock, & Briere, 1996). Sexual abuse, therefore, affects a large percentage of children, who are at considerable risk for impaired mental health and adjustment in both childhood and adulthood.

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Certain factors pose risks for more serious consequences. These include aspects of the abuse (sexual abuse within the family, ongoing and multiple episodes, the use of force, methods used to gain child secrecy, and multiple perpetrators), legal system involvement (a case that results in acquittal), and maternal response to the sexual abuse (lack of maternal support and maternal use of avoidant coping mechanisms) (Wolfe, 1998; Wolfe & Birt, 1997). Many of these variables are not subject to change, such as aspects of abuse that have already occurred, but maternal support is one area that can be affected by intervention, with significant consequences for both short- and long-term adjustment (Adams-Tucker, 1982; Conte & Schuerman, 1987; Edwards & Alexander, 1992; Everson, Hunter, Runyon, Edelsohn, & Coulter, 1989; Gold, 1986; Nash, Hulsey, Sexton, Harralson, & Lambert, 1993; Peters, 1988; Tufts New England Medical Center, Division of Child Psychiatry, 1984; Wind & Silvern, 1994; Wyatt & Mickey, 1987). Mothers may also suffer a great deal of distress at learning of their children's sexual abuse, and their negative reactions may impede children's recovery (see Corcoran, 1998).

Intervention may be helpful for caregivers to assist them in coping with stressors associated with sexual abuse of their child, such as separation from a partner, loss of social support, and being involved in legal proceedings. Learning about abuse dynamics can also reduce distress (Deblinger & Heflin, 1996). For instance, caregivers are

often upset because they did not learn of the abuse sooner, but they can be told that perpetrators often involve their victims in secrecy. In addition, treatment can assist mothers in learning about how to identify and handle symptoms in their children and to respond appropriately to children's questions and concerns.

For these reasons, treatment involving the nonoffending caregiver, as well as the child victim, has been developed (i.e., Cohen & Mannarino, 1993; Deblinger & Heflin, 1996). These treatments are typically cognitive-behavioral in their theoretical orientation and involve separate sessions for the caregiver and the child, although at the latter part of treatment, conjoint sessions are often involved.

Reviews of this treatment outcome literature indicate support for the parent-involved treatments. For instance, Putnam (2003) claimed that they "are currently the best-documented, effective treatments" for sexual abuse in children. However, Putnam further stated that certain symptoms, including aggression and sexualized behavior, were difficult to affect by these interventions. A review of studies conducted before 2000 also touted abuse-specific cognitive-behavioral therapy over comparison group conditions (Saywitz, Mannarino, Berliner, & Cohen, 2000). These authors concluded that training parents in behavioral techniques is necessary to reduce externalizing symptoms and sexual acting out in children. Another review classified abuse-focused, parent-involved, cognitive-behavioral therapy as the only "well-supported, efficacious treatment" for sexually abused children (Saunders, Berliner, & Hanson, 2003). Key components of well-supported, efficacious treatments include that they offer manualized treatment that has produced superior results over comparison conditions in at least two randomized controlled trails.

A final review summarized outcomes of parent-involved sexual abuse treatment by developmental stage (Corcoran, 2004). For the preschool-age child and their parents, cognitive-behavioral treatment of parents and children offered benefits in many domains for both parents and children over nondirective supportive comparison treatment. Although mixed findings were found for the effect of treatment on children's externalizing symptoms, children's sexualized behavior, a particularly intractable symptom area, was affected positively. Just as for the results of sexually abused preschool children, cognitive-behavioral parent-involved treatment showed clear benefits over the comparison conditions in studies for school-age children across multiple domains of child functioning. However, posttraumatic stress disorder (PTSD) was one domain in which child-only treatment alone seemed to confer benefits (Deblinger, Lippmann, & Steer, 1996; King et al., 2000).

To summarize the narrative reviews, parent-involved treatment for sexual abuse in children has been regarded as one, if not the best, available treatment for child sexual victimization. Narrative reviews are mixed as to whether aggressive/externalizing and sexual acting out in children can be affected by these interventions. The findings of these narrative reviews are complicated by the fact that outcomes in these studies are often many, including PTSD, other anxiety symptoms, depression, behavior problems, and sexual acting out, and by the use of multiple measures to assess these outcomes. For these reasons, it is difficult to summarize findings across studies at the different periods at which data are collected.

Narrative reviews are subject to certain limitations, namely their reliance on statistical significance. This "vote count" method—counting a study's findings as positive if statistically significant results are reported—is vulnerable to certain bias. A major limitation of vote counting is its reliance on tests of statistical significance in the primary studies. Significance tests are affected by sample size. In studies with very large samples, clinically insignificant differences will be statistically significant. Significance tests (p values) do not tell us the strength or magnitude of the effect, so vote counting may tally positive scores that do not really matter (clinically insignificant effects in large studies). Conversely, studies based on small samples often lack the statistical power needed to detect meaningful effects. Thus, clinically significant effects can be missed when small studies are combined with vote counting.

During the review process of this article, a systematic review of cognitive-behavioral treatment for child sexual abuse victims was published (Macdonald, Higgins, & Ramchandani, 2006). In all, 10 studies were located that met the inclusion criteria. Although some support was provided for the effectiveness of cognitive-behavioral treatment, the conclusion was tempered by the quality of the primary studies.

Because we were interested in the specific impact of parent involvement in child sexual abuse treatment, not necessarily the theoretical orientation of this treatment and because of the limitations inherent in the narrative reviews conducted in this area, a meta-analysis was undertaken of this literature.

METHODOLOGY FOR META-ANALYSIS

The main purpose of the study was to conduct a meta-analysis to determine how parent-involved treatment of child sexual abuse victims affected child adjustment in terms of the following symptom areas when compared to another type of treatment or a control group: internalizing

symptoms, externalizing symptoms, sexualized behaviors, and/or PTSD. A secondary purpose was to determine the influence of participant-, intervention-, and design-related variables on symptom domain effect sizes.

Inclusion criteria for the meta-analysis included the following: (a) Sexual abuse was the reason for treatment referral; (b) treatment was provided to children rather than adult survivors of sexual abuse; (c) parents had to be involved in the treatment, although the child's sexual abuse was a focus of the intervention; (d) studies had to assess the result of treatment on child internalizing symptoms (e.g., depression, anxiety), externalizing (behavior problems), sexual behavior problems, or PTSD; and (e) comparison or control group designs were required. Although randomization to treatment and control conditions was not necessary, single-group pretest, posttest designs were excluded, and (f) authors of studies had to provide the necessary statistical information to calculate effect sizes.

Databases were searched from 1980 to summer 2005 and included the following: CINAHL, Infotrac, Medline, Psychinfo, Social Sciences Index, and Social Work Research and Abstracts. Search terms included child sexual abuse (e.g., victimization, molestation), treatment (e.g., therapy, intervention), and outcome (e.g., research, evaluation). The National Clearinghouse of Abuse and Neglect and the four first authors of the primary parent-involved studies were also contacted for their knowledge of other studies meeting our criteria. Only one of the study authors responded to e-mail inquiry, and no new studies were added as a result of these contacts. Furthermore, the reference lists of relevant literature reviews were searched for possible studies. Most recently, the reference list for the systematic review on cognitive-behavioral treatment of child sexual abuse (Macdonald et al., 2006) was searched for additional studies, but no new studies were found.

Although the exact number of titles and abstracts screened was not tracked, common reasons for exclusion were the following: only single group pretest/posttest group designs were used, parents were not involved in the child's treatment, the study outcomes studies were not one of the four outcomes of interest, and statistical information necessary to calculate an effect size was not present. For reasons for exclusion of particular studies, please contact the first author. When all studies were screened, seven outcome studies became the subject of this meta-analysis. Because of the few number of studies, moderator analysis was precluded.

Coding

Studies were coded both by the principle investigator and a team of two trained master's in social work students, who were taking an advanced research course. In addition to statistical data, design-, participant-, and treatment-related variables were also coded. Design-related variables included the research design, the type of comparison/control group employed, the presence of follow-up and its length, measures used to assess internalizing and externalizing symptoms and sexual behavior problems, and quality rating. Participant-related variables included race, gender, socioeconomic status, and mean age of the children treated. Treatment-related variables involved the theoretical orientation of the treatment, the length of the intervention in terms of session length and number of sessions, and the modality employed. Any discrepancies in coding were discussed to consensus.

Quality aspects of studies included the following: type of randomization method, the extent to which participant and assessment blinding occurred, the extent of attrition, and whether intention-to-treat analyses were conducted.

Data Analysis

To compare the effectiveness of interventions across studies, the effect size measure known as Hedges's (1981) *g*, a standardized mean effect size formula that corrects for small sample size, was computed using Comprehensive Meta-Analysis (Borenstein, Hedges, Higgins, & Rothstein, 2005). Because all data were continuous, the same effect size formula was used throughout.

Estimates of effect sizes are expressed as fractions or digits and are called "point estimates." They are estimates of the unknown population parameter, the effect size of interest. A single number may not provide a reasonably close estimate of the population because of random error and other factors. To get a more comprehensive estimate of the effect size, the range of probabilities, including the true population effect size, is provided through the construction of a confidence interval. The confidence interval incorporates a set of possible values that includes, with specified probability, the true mean. Three pieces of information are necessary to construct a confidence interval: the effect size (ES), the standard error (*SE*) of the effect size, and the *z* score, the value of which will depend on the selected level of confidence, such as .05. The confidence interval is created around X, the population effect size being estimated. The length of the interval on either side of X is obtained using the formula: $ES - Z^{*}se \leq X \leq ES + Z$. If the confidence interval includes the "null effect," the effect size is not statistically significant. If the confidence interval does not include the "null effect" value, then the effect size is significant at the selected level of confidence.

Estimation of the mean effect size requires making a choice between two types of assumptions with respect

to the error structure: fixed effect and random effect modeling. Fixed and random effect modeling are based on two different conceptual approaches to meta-analysis, which lead to different calculations of the overall mean effect size. Under the fixed effect approach, the overall mean effect size is obtained while making allowances for the random error within studies. In the fixed effect modeling, the variation in effect size between studies is ignored; the assumption is that all studies estimate the same effect and that study effects vary only because of random variation between research participants (Raudenbush, 1994). As a result, the weights assigned to the studies depend on the sampling error within the studies.

Under the random effect model, the combined effect is an average of a number of effect size estimates of different populations. Here, the overall effect size incorporates information about sampling error within studies as well as the variation in effect size between studies. Even small studies provide unique information about population effect size. This information is included to arrive at an overall mean effect size.

The random effect model takes into account more sources of variation in effect sizes than fixed effect models. As a result, the confidence intervals of the overall mean effect size in the random effect model are likely to be wider than the same under the fixed effect assumption. The conceptual basis for random effect modeling suggests that both within-study error and between-study variance need to be taken into account. The total variation in random effect modeling is then apportioned into within-study error and between-study variance. When between-studies variance is small, the total variation is assigned to within-study error. Under this assumption, there is no difference in the overall effect size derived from either the fixed effect or the random effect modeling approach. However, when between-study variance is assumed to be present, the random effect model is an appropriate choice to estimate the overall effect size.

In this study, the mean effect sizes were calculated under both assumptions separately. Under the fixed effect model, if the pooled variance estimate is large, greater than 0, the assumption that the only source of variation in true effect sizes is due to sampling error may not be appropriate. Thus, in meta-analysis, random effect models are used to explain heterogeneity among effect sizes beyond the variations related to fixed effects.

In the meta-analysis, the extent of homogeneity among effect sizes related to each of the four outcomes is indicated by the Q statistic. The total heterogeneity, QT, is tested against a chi-squared distribution with n-1 degrees of freedom (Gurevitch & Hedges, 1993s). In the event of sufficient numbers of studies, moderator analysis would be conducted to determine if participant-, intervention-, or design-related variables were associated with varying effect sizes.

RESULTS

The treatment-, participant-, and design-related information of studies included in this meta-analysis are portrayed in Table 1. Assessment of methodological qualities of studies is displayed in Table 2. Because only two studies used a no-treatment control group and all relied on comparison groups, the decision was made to calculate effect sizes on only the treatment versus comparison groups so as to be consistent. This also observes the recommendation that independent effect sizes should be calculated from each study in a meta-analysis (Hedges & Olkin, 1985). When multiple comparison groups were employed in studies, the decision was made to use only the child-oriented treatment condition, as that was seen as the most widely used approach in practice settings. Finally, in the case of multiple follow-up periods (Deblinger et al., 1996), the decision was made to take the first follow-up period.

Results of the meta-analysis will be provided separately for each outcome. Although results are presented for both posttest and follow-up effect sizes, the reader should note that the follow-up effect sizes are based on very few studies and should be viewed with caution as a result.

Internalizing Symptoms

Six of the seven studies used the Child Behavior Checklist Internalizing and Externalizing scales (Achenbach & Edelbrock, 1991). Although studies employed other measures for depression, anxiety, and behavior problems, these were few and measures used were not consistent across studies. Therefore, to maintain homogeneity, only the Child Behavior Checklist scales were used to represent internalizing and externalizing symptoms; in this way, comparisons across studies could be made more easily.

Table 3 presents the effect sizes and variances for all the outcomes related to "internalizing." Figure 1 presents the confidence intervals for each of these effect sizes. At posttest, two of the six effect sizes are moderate, with values greater than 0.5 and less than 0.8 (Cohen, 1988), with the treatment effect in the expected direction. However, all but two of the confidence intervals for these effect sizes contain the null effect size. Table 4 presents the mean effect size under the fixed and random effect assumptions. The p value of the Q statistic, with a magnitude of 3.45 and 5 degrees of freedom,

TABLE 1: Treatment and Design Information on Meta-Analysis Studies

Description of Treatment	Sample	Design	Outcomes
Celano, Hazzard, Webb, & McCall (1996) Eight 1-hr sessions divided between caregiver and child; 2 to 3 conjoint	*N* = 32 sexually abused girls (ages 8 to 13) and their nonoffending caregivers 75% African American	Experimental, randomization to treatment vs. treatment-as-usual usual	CBCL Internalizing CBCL Externalizing CBCL PTSD CITES-R PTSD
Cohen & Mannarino (1996, 1997) Cognitive-behavioral 12 sessions, individual with both parent and child	*N* = 67 children (*M* age = 4.68) and caregivers 50% male, 50% female; 50% White, 50% African American	Experimental, randomization to cognitive behavioral or individual supportive therapy, 1-year follow-up	CBCL Internalizing CBCL Externalizing Child Sexual Behavior Inventory
Cohen & Mannarino (1998a); Cohen, Mannarino, & Knudsen (2005)a Cognitive-behavioral 12 sessions, individual with both parent and child	*N* = 67 children (*M* age = 11.08) and caregivers 69% female; 59% White, 37% African American	Experimental, randomization to cognitive behavioral or individual supportive therapy, 1-year follow-upb	CBCL Internalizing CBCL Externalizing Child Sexual Behavior Inventory
Cohen et al. (2004) 12 weekly cognitive-behavioral individual sessions for parent and child. In three sessions, a joint 30-min parent-child session was held.	*N* = 180 children (*M* age = 10.76) and caregivers 79% female; 60% White, 28% African American	Experimental, randomization to parent-child CBT or child supportive therapy	CBCL Internalizing CBCL Externalizing Child Sexual Behavior Inventory
Deblinger et al. (1996); Deblinger, Steer, & Lippmann (1999) 12 individual cognitive-behavioral sessions for both parent and child separately; in latter stage of therapy, cojoint sessions	*N* = 90 children (ages 7 to 13) 83% female; 72% White, 20% African American	Experimental, randomization to three experimental conditions: parent-only, child-only, parent and child, and community control, follow-up (6, 12, and 24 months)	CBCL Internalizing CBCL Externalizing PTSD section of Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children-Epidemiologic version
Deblinger, Stauffer, & Steer (2001) Cognitive-behavioral group therapy for both child and parent; 8 sessions; for cognitive-behavioral condition, 15 minute parent-child activity per week.	*N* = 44 children (ages 2 to 8) and caretakers 61% female; 64% White, 21% African American	Experimental, randomization to cognitive behavioral or supportive therapies, 3-month follow-up	Child Sexual Behavior Inventory PTSD section of Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children-Epidemiologic version
King et al. (2000) Cognitive-behavioral 20 individual parent and child sessions focusing on posttraumatic symptoms	*N* = 36 children (*M* age = 11.4) 69% female, 31% male	Experimental design, randomized to child alone cognitive-behavioral treatment, family cognitive-behavioral treatment, or waiting-list control, 12-week follow-up	CBCL Internalizing CBCL Externalizing CBCL PTSD PTSD section of the child version of the Anxiety Disorders Interview Schedule for *DSM-IV*

NOTE: CITES-R = Children's Impact of Traumatic Events Scale–Revised; CBT = cognitive behavioral therapy; CBCL = Child Behavior Checklist; PTSD = posttraumatic stress disorder; DSM-IV = *Diagnostic and Statistical Manual of Mental Disorders* (4th ed.; American Psychiatric Association, 1994).
a. The reason more than one citation is listed for some studies is that investigators reported results for posttest and follow-up in separate publications.
b. The follow-up study (Cohen et al., 2005) was not included in the meta-analysis because the analysis was done on intent-to-treat rather than completers of treatment.

is not significant at the .05 level, suggesting the absence of significant heterogeneity among the effect sizes.

The mean effect size under the random effect model assumption is .41, with a confidence interval ranging from .21 to .61, suggesting a small treatment effect on internalizing outcome. The mean effect size at follow-up (represented by only two studies) is small, nearly .35. However, the confidence interval is large, with the null effect size well within the range of the confidence interval (see Figure 2).

Externalizing Symptoms

Six studies reported externalizing outcomes at posttest, whereas three assessed these outcomes at follow-up. Table 5 presents the effect sizes and variances for all the externalizing outcomes. Figure 3 presents the confidence intervals for each of these effect sizes. Most of the effect sizes are small. The mean effect size presented in Table 4 is small, nearly .32, and has a confidence interval that does not contain the null effect size (see Figure 3). The *p* value

TABLE 2: Methodological Quality of Studies[a]

Study ID	Randomization	Participant Blinding	Assessment Blinding	Attrition	Intention-To-Treat Analyses
Celano et al. (1996)	Method not reported	Not reported	A psychiatrist or psychologist blind to the treatment condition rated the child's overall psychosocial functioning based on child and caretaker interviews. Standardized measures were administered by a clinician not involved in the child's treatment, but not reported as blind to the treatment condition.	Two participants dropped out before treatment began. It is not clear whether they had already been randomized. Of the 15 who dropped out after treatment began, 10 dropped out of the experimental group and 5 from the control group.	Not reported
Cohen & Mannarino (1996, 1997)	Efron's biased coin toss	Not reported	No. All but one of the child outcome measures was completed by the child's parent.	Thirteen of the 86 participants dropped out during treatment. A further 6 were removed because of persistent sexually inappropriate behavior. No information is provided about the distribution of these dropouts and exclusions. At follow-up, data on only 43 of the original participants are presented. No information is provided about the reasons for these subsequent losses of data (other than the "kinds" of reason) or their distribution across groups.	Not reported
Cohen & Mannarino (1998b)	Efron's biased coin toss/ random numbers series generated by computer	Not reported	2004 paper reports that the evaluator conducting initial and follow-up assessments was blind to treatment condition or assignment.	Four participants dropped out between assessment and intervention. Ten participants dropped out after completing three or fewer sessions. Ten dropped out after completing between four and eight sessions. Nine were removed for persistent contact sexualized behaviors. Eleven children dropped out of the experimental group, 22 from the control group. Larger numbers of children were removed from the control group than from the experimental group for ongoing sexual behaviors toward other children (7 compared to 2).	Yes
Deblinger et al. (1999)	Method not reported. Participants randomized after initial assessment.	Not reported	Not reported. Therapists were involved in the collection of baseline data.	Ten participants dropped out of the study during treatment. The dropout appears to be three participants from the waitlist control and two of the experimental groups and one from the child-only experimental group. Missing data increased over time, but no information given on the total numbers or their distribution across groups.	Not reported
King et al. (2000)	Method not reported. Participants randomized after pretreatment assessment.	Not reported	Assessors not blinded. Assessments conducted by therapists.	Between recruitment and treatment 10 families lost out of 32 eligible families referred. During treatment Experimental groups each lost three participants. Community control group lost two participants.	Yes

(continued)

TABLE 2: (continued)

Study ID	Randomization	Participant Blinding	Assessment Blinding	Attrition	Intention-To-Treat Analyses
Cohen et al. (2004)	Method not reported	Not reported	All therapists trained in both modalities, therefore assessors were blind to which therapy had been used in any particular case.	Twenty-six of the 229 children originally randomized dropped out of the study. Five (2%) never returned for treatment. Eight (3%) left after attending one session. Thirteen (6%) left after attending only two sessions.	Yes
Deblinger et al. (2001)	Therapeutic method was randomly determined by computer program prior to the initiation of a group.	Not reported	All therapists were trained in both approaches used	Sixty-seven mothers volunteered for the project. Four did not return for assignment. Five left after attending one session. Four left after attending two sessions. These 13 were treated as dropouts. Of the 54 who completed at least three group sessions (and were therefore included in the analyses) only 44 completed both posttest and 3-month follow-up evaluations. One mother failed to complete the posttest evaluation and a further nine failed to complete either. The 2001 paper reports data on these 44.	No

TABLE 3: Effect Sizes (Hedge's *d*) for Internalizing Outcomes at Posttest

	Posttest			Follow-Up		
Study	Effect	Variance	Probability	Effect	Variance	Probability
King et al. (2000)	.6076	.2208	.177	.474	.217	.287
Deblinger et al. (1996)	.2071	.0876	.478			
Cohen & Mannarino (1998a, 1998b)	.2828	.0868	.329			
Celano et al. (1996)	.0913	.1256	.792			
Cohen & Mannarino (1996)	.7423	.0655	.003	.299	.103	.342
Cohen et al. (2004)	.4181	.0227	.005			

TABLE 4: Mean Effect Size and Measure of Heterogeneity for Effect Sizes

			95% CI					
Variable (Time Frame)	Type	Point Estimate	Lower	Upper	Probability	Heterogeneity (Q Value)	*df*	Probability
Internalizing (posttest)	Fixed	.409	.210	.608	.000	3.449	5	.631
	Random	.409	.210	.608	.000			
Internalizing (follow-up)	Fixed	.358	–.147	.863	.165	0.102	2	.165
	Random	.358	–.147	.863	.165			
Externalizing (posttest)	Fixed	.323	.125	.521	.001	2.097	5	.837
	Random	.323	.125	.521	.001			
Externalizing (follow-up)	Fixed	.187	–.205	.578	.350	0.430	2	.807
	Random	.187	–.205	.578	.350			
Sexual behaviors (posttest)[a]	Fixed	.310	.096	.524	.005	1.350	3	.717
	Random	.310	.096	.524	.005			
PTSD (posttest)	Fixed	.365	.144	.545	.001	3.450	4	.485
	Random	.365	.096	.545	.001			
PTSD (follow-up)	Fixed	.249	–.132	.631	.201	0.213	2	.899
	Random	.249	–.132	.631	.201			

NOTE: PTSD = posttraumatic stress disorder; CI = confidence interval.
a. A follow-up effect size is not provided for sexual behaviors because results were only based on one study.

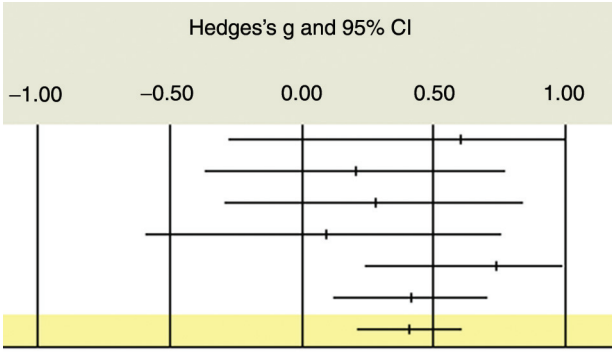

Figure 1: Confidence Intervals (CI) for Internalizing Outcomes Effect Sizes at Posttest

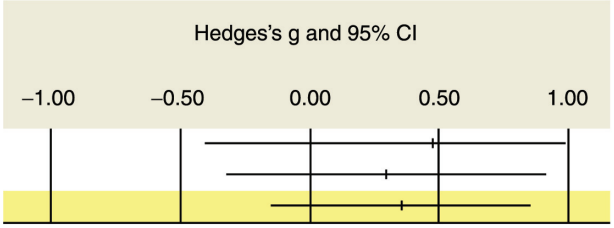

Figure 2: Confidence Intervals (CIs) for Internalizing Outcomes Effect Sizes at Follow-Up

of the Q statistic, with a magnitude of 2.10 and 5 degrees of freedom, is not significant at the .05 level, suggesting the absence of significant heterogeneity among the effect sizes. The mean effect size at follow-up is small, and the null effect size is well within the range of the confidence interval (see Figure 4).

Sexual Behaviors

Child sexual behaviors were assessed less frequently in studies; only four of the studies researched this outcome. However, all used the Child Sexual Behavior Inventory (Friedrich et al., 1992). Only one study followed up with this outcome over time. Effect sizes related to sexual behavior are presented in Table 6. Two of the effect size values, .54 and .38, are medium and small, and the remaining two are small effects. Two of the four effects, as presented in Figure 5, have confidence intervals that do not contain the null effect. The mean effect size, under the fixed effect assumption, is .31 (see Table 4). The effect is small, and its confidence interval does not contain the null effect. These results suggest that, very likely, the treatment effect on behavioral outcomes is significant. The *p* value of the Q statistic, with a magnitude of 1.35 and 3 degrees of freedom, is not significant at the .05 level, suggesting the absence of significant heterogeneity among the effect sizes.

Figure 6 presents the effect size for the only sexualized behavior outcome at the follow-up stage. The effect size (Hedges *d*) is .445, with a variance of .022 and a *p* value of .003 (see Table 6). The lower and upper limit of the confidence intervals are .152 and .739, respectively.

PTSD

Five studies reported PTSD outcomes at posttest, whereas only three assessed this outcome at the follow-up stage. Table 7 presents these effect sizes and variances, and Figure 7 demonstrates the confidence intervals for each of these effect sizes. At posttest, most of the effect sizes are small. Only one of the six effect sizes, that with the value .526, does not contain the null effect size well within the confidence interval. The mean effect size as presented in Table 4 is small, nearly .365 under the fixed

TABLE 5: Effect Sizes (Hedge's *d*) for Externalizing Outcomes

	Posttest			Follow-Up		
Study	Effect	Variance	Probability	Effect	Variance	Probability
Externalizing						
King et al. (2000)	.163	.212	.710	.351	.196	.428
Deblinger et al. (1996)	.429	.089	.144	.030	.101	.924
Cohen & Mannarino (1998a, 1998b)	.098	.086	.735			
Celano et al. (1996)	.701	.127	.049			
Cohen & Mannarino (1996)	.377	.061	.127	.257	.099	.413
Cohen et al. (2004)	.288	.022	.050			

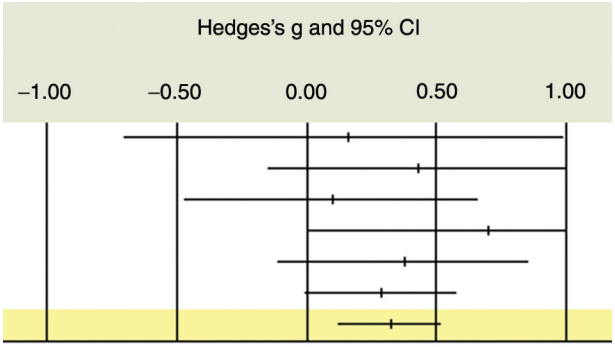

Figure 3: Confidence Intervals (CIs) for Externalizing Outcomes Effect Sizes at Posttest

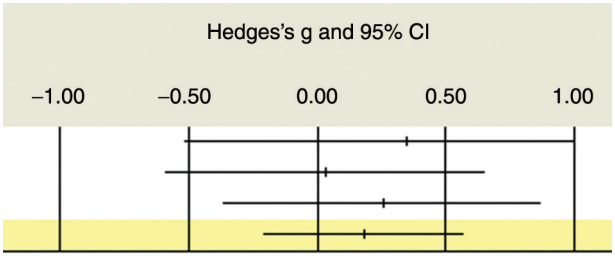

Figure 4: Confidence Intervals (CIs) for Externalizing Outcomes Effect Sizes at Follow-Up

effect assumption. The *p* value of the Q statistic, with a magnitude 3.45 and 4 degrees of freedom, is not significant at the .05 level, suggesting the absence of significant heterogeneity among the effect sizes.

At follow-up, all three effect sizes contain the null effect size well within their confidence intervals (see Figure 8). The mean effect size (see Table 4) is small, and the null effect size is comfortably within the range of the confidence interval of the mean effect size.

Fail-Safe *N*

The fail safe *N* is the number of unpublished studies that would have to be added to the meta-analysis to move the results from significance to nonsignificance. The fail safe *N* using the Rosenthal method is approximately 16 for the internalizing symptoms domain, 10 for the externalizing outcome, 5 for the sexualized behavior outcome, and 4 for the PTSD outcomes. These fail safe *N*s are not necessarily large and call for guarded caution with respect to the findings of this meta-analysis.

DISCUSSION AND APPLICATION TO SOCIAL WORK PRACTICE

In general, the results of this meta-analysis suggest that, at posttest, parent-involved treatment of child sexual abuse victims have small effects on child adjustment over various types of comparison treatment. Internalizing outcomes show the strongest effects, followed by PTSD, sexual behaviors, and externalizing behaviors. There was question in the narrative reviews about whether parent-involved treatment affected externalizing symptoms, sexual acting out, and PTSD symptoms over child treatment. It is interesting that effects were quite similar across symptom domains in terms of their being categorized as small, according to Cohen (1988). These findings can be viewed important as most of the comparison conditions involve intervention (usually child treatment). Indeed, according to Lipsey (1998), an effect size of about .20 has practical significance in intervention research.

At follow-up, treatment effects on the four categories of outcomes were smaller still (except for sexualized behaviors), and the confidence interval for the mean effect size for each symptom domain contained the null effect. It appears, therefore, that the effects seen at posttest tend to diminish by the follow-up period. It must be recognized that far fewer studies assessed outcome at the follow-up period than at posttest (and for sexual behaviors, there was only one study), so these

TABLE 6: Effect Sizes (Hedge's *d*) for Sexual Behaviors Outcomes

	Posttest			Follow-Up		
Study	Effect	Variance	Probability	Effect	Variance	Probability
Deblinger et al. (2001)	.383	.090	.201			
Cohen & Mannarino (1998a, 1998b)	.230	.084	.426			
Cohen & Mannarino (1996)	.548	.062	.028	.445	.022	.003
Cohen et al. (2004)	.227	.022	.128			

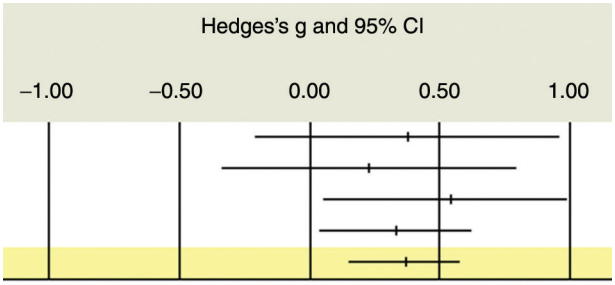

Figure 5: Confidence Intervals (CIs) for Sexual Behavior Outcomes Effect Sizes at Posttest

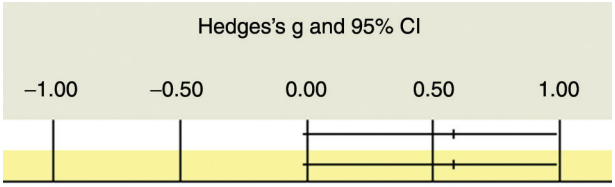

Figure 6: Confidence Intervals (CIs) for Sexual Behavior Outcomes Effect Sizes at Follow-Up

conclusions do not represent all the studies assessed at posttest. The research recommendation, therefore, is that parent-involved treatment outcome studies should routinely assess participants at a follow-up period. Finally, it should be noted that the treatments represented here are "brief" in nature—8 to 20 sessions. This may indicate that treatments may have to be of longer duration or involve maintenance or "booster" sessions after acute treatment so that treatment gains are sustained and strengthened.

Limitations

In addition to the few studies that assessed follow-up information, other limitations of this study are those inherent to meta-analysis. Although every attempt was made to gather all studies that met the criteria, some publications may have been inadvertently overlooked. Another problem was the lack of comparison/control group studies to include in the meta-analysis. Indeed, so few studies ended up in the meta-analysis that moderator analysis was precluded. Moderator analysis could reveal potentially useful information, such as the demographic characteristics of the clients or the types of interventions that contributed to better outcomes.

Another limitation involved the fact that the alternative treatments involved in the effect size calculations varied across studies. Some of the alternative treatments comprised other types of parent involvement (i.e., treatment as usual), and some used child-only treatment.

For the internalizing, externalizing, and sexualized behaviors outcomes, results were based on parent report. Although there are good reasons to rely on parent report for certain behaviors, particularly externalizing symptoms (Loeber, Loeber, Wei, Farrington, & Wikström, 2002) and sexualized behaviors (Friedrich et al., 1992), parent reports of internalizing symptoms may not be as accurate (Birmaher et al., 1996). Although the internalizing outcome had one of the highest effect sizes overall, this result may be inflated by parents' tendencies to see their children as less depressed, for instance, than perhaps they are.

Part of the rationale for including parents in treatment, as discussed earlier, is because of the distress from which parents suffer from sexual abuse disclosure. At this point, not all studies focused on how parents are affected by the intervention; in those studies that did examine the result of treatment on parent functioning, there was a lack of consistency in measures across studies. Therefore, currently,

TABLE 7: Effect Sizes (Hedge's *d*) for PTSD Outcomes

	Posttest			Follow-Up		
Study	Effect	Variance	Probability	Effect	Variance	Probability
King et al. (2000)	.362[a]	.206	.425	.254	.204	.574
Deblinger et al. (1996)	.037	.084	.900	.144	.098	.643
Deblinger et al. (2001)	.064	.088	.828	.343	.089	.250
Celano et al. (1996)	.403[a]	.139	.280			
Cohen et al. (2004)	.526	.012	.001			

NOTE: PTSD = posttraumatic stress disorder.
a. Denotes the average of more than one PTSD outcome measure.

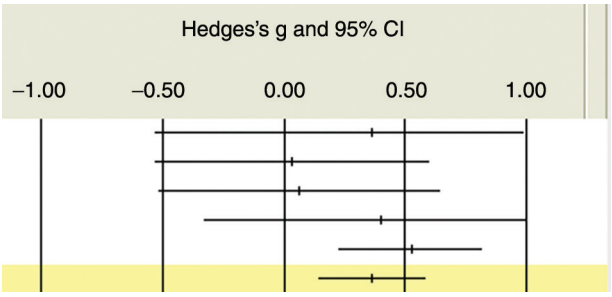

Figure 7: Confidence Intervals (CIs) for Posttraumatic Stress Disorder (PTSD) Effect Sizes Related to PTSD at Posttest

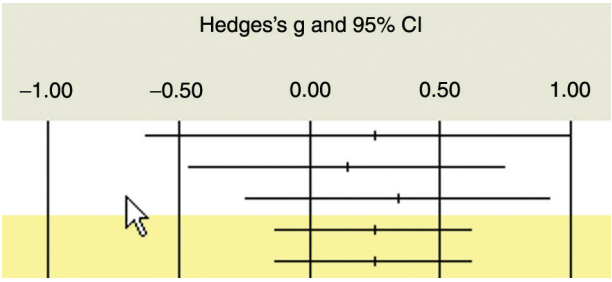

Figure 8: Confidence Intervals (CIs) for Posttraumatic Stress Disorder (PTSD) Effect Sizes at Follow-Up

there are an insufficient number of studies to quantitatively summarize the effect on parents. Researchers in this area might also do well to select parent measures that can be completed across studies to gain an assessment of the overall impact to parents from their involvement in treatment.

One area of interest is that two research teams (Cohen and Mannarino and Deblinger and colleagues) represent five of the seven studies in this meta-analysis. Although careful analysis of the studies determined that they represented nonoverlapping samples, whether their results will be replicated with other investigators will depend on future research.

Despite the limitations, this meta-analysis contributes to the literature on sexual abuse treatment by quantifying the effects of parent-involved treatment on the many domains of functioning that sexual abuse may affect. Overall, parent-involved treatment confers some advantage over comparison conditions (typically child-only treatment) at posttest. This finding extends across all domains studied: internalizing symptoms, externalizing symptoms, sexualized behaviors, and PTSD. Therefore, the question in the literature about the types of problems that parent-involved treatment may affect appears moot; it appears that all outcomes are affected at about an equal amount (a small effect). At follow-up, effect sizes tended to diminish, although more follow-up studies are needed to replicate these meta-analytic results.

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